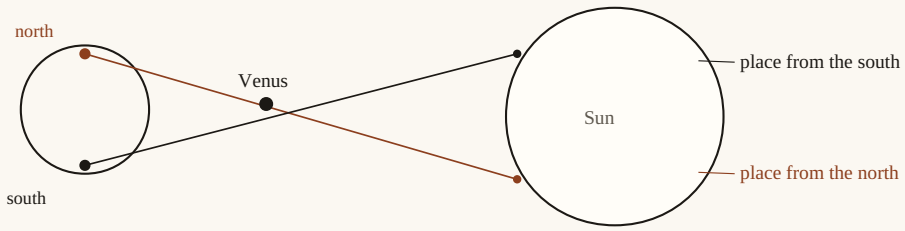
FARIA · 7 NOVEMBER 2026 · A LESSON FOR JEFF FRANZEN

The Venus transits: a worldwide ruler

Kepler's clock gives the planets in units of Earth's orbit. A transit of Venus can turn that ratio into a length, if two stations far apart time the same event. What they compare is a differential parallax: Venus, being nearer, shifts on the Sun more than the Sun itself shifts. The full story of the ladder, including why 23 arcseconds is a one-radius illustration and not a ceiling, is the Venus-to-Cepheid lesson. This page is the human campaign.

What the stations compare

Schematic. Angles exaggerated. Not a traced 1769 track.



Differential parallax is about (1/0.277 - 1) times the solar parallax, about 2.61 times it.

One Earth radius, side-on, is about 23 arcseconds if the solar parallax is the modern 8.79.

That length is an illustration, not a ceiling. Stations can stand up to about a diameter apart.

Halley compared full internal-contact durations. One contact, timed in absolute time, needs longitude.

The black drop is mostly blur and the Sun's dim limb, not Venus's air by itself.

Two stations and the differential shift of Venus on the Sun

Teaching geometry, not a 1769 tracing. Halley compared durations of the internal contacts. A single contact timed on an absolute clock needs the longitude. The black drop is mostly blur and limb darkening.

Jeremiah Horrocks and William Crabtree, observing separately, saw the transit of 1639. Horrocks argued from Venus's small apparent disk, and from a model of sizes, that the solar parallax was not above 14 arcseconds, and the Sun at least about 15,000 Earth radii. That is not a two-station parallax. The book came out after his death, in 1662. Converted with modern constants, 14 arcseconds is about 58.4 million miles and 15,000 Earth radii about 59.4 million miles. Those mile figures are our conversion, not his words.

In 1761 Mason and Dixon were sent toward Bencoolen. Their ship, the Seahorse, was attacked in January. Mason's letter counts eleven killed and thirty-seven wounded. After repairs they observed from the Cape, and only the end of the transit, after sunrise on 6 June. Cloud early, then a clearing for the contacts. Do not promote that into a cloudless day. James Short compared the Cape with fifteen European stations. He was not their exclusive partner. His 8.47 and 8.52 arcseconds are for the day of the transit, not the mean distance. The mean-distance figure near 8.65 converts to about

94.5 million miles, roughly 1.7 percent high. A later Short value, 8.69, converts to about 94.1 million miles. Again, the miles are a modern conversion.

There was no one official answer. Pingre's 1765 reduction from Rodrigues, 10.25 arcseconds, converts to about 79.8 million miles, about 14 percent low. Hornsby, using 1769 and publishing in 1771-72, got a mean near 8.78 arcseconds, about 93.1 million miles, roughly 0.16 percent above today's 92.956 million miles. The astronomical unit is now defined as exactly 149,597,870,700 metres. Encke in 1824 combined both transits and got 8.5776 arcseconds, about 95.3 million miles, 2.5 percent high. That combination is not the 1769 result alone.

Some observers got little or nothing. Le Gentil, kept from Pondicherry by war in 1761, watched from a moving ship and the record was unusable. In 1769 a cloud hid Pondicherry again. Maskelyne at St Helena could not send a complete observation. Pingre at Rodrigues saw the Sun rise in thick cloud. At Greenwich, Bliss and Green missed the start for cloud.

In 1769 Cook and Green had a clear 3 June at Tahiti, after the quadrant was stolen on 2 May and recovered damaged but repairable. Green died of dysentery on the way home, on 29 January 1771. Chappe observed successfully at San Jose del Cabo and died on 1 August 1769 in an epidemic. The disease has been given more than one name, so this page says epidemic. Wales and Dymond wintered at Hudson Bay and did observe. Their story is not a stopped clock. Mason's share of the Cape work is the next lesson.

The illustrated lesson is download the PDF.

Sources

- Andreas Verdun, "The Determination of the Solar Parallax from Transits of Venus in the 18th Century," Archives des Sciences 57 (2004): https://www.unige.ch/sphn/Publications/ArchivesSciences/AdS%202004-2015/AdS%202004%20Vol%2057%20Fasc%201/45-68%20Verdun.pdf
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Key facts

- Horrocks in 1639 set an upper bound near 14 arcseconds from Venus's apparent size, not from a second station. Publication was posthumous, 1662.
- Mason and Dixon reached the Cape after the Seahorse fight (11 dead, 37 wounded) and timed the end of the 6 June 1761 transit.
- Short, Hornsby, Pingre, and Encke published different parallaxes. Converted miles on this page are modern arithmetic, not their sentences.
- The modern au is exactly 149,597,870,700 metres, about 92.956 million miles. No 1761 or 1769 paper printed that definition.